Special right triangles

1 The side opposite the angle measuring 30°.

2 1 unit

3

a $b=\sqrt{3}$

b $\theta=30^\circ$

4

a

a	1	2	3	4	5
b	1	2	3	4	5
c	$\sqrt{2}$	$2\sqrt{2}$	$3\sqrt{2}$	$4\sqrt{2}$	$5\sqrt{2}$

b x units

c $\sqrt{2}$ units

d The two shortest sides have the same length and the length of the hypotenuse is $\sqrt{2}$ times larger than either of the other side lengths.

5

a

a	1	2	3	4	5
b	$\sqrt{3}$	$2\sqrt{3}$	$3\sqrt{3}$	$4\sqrt{3}$	$5\sqrt{3}$
c	2	4	6	8	10

b $x\sqrt{3}$ units

6

a

i $a=15$

ii $c=15\sqrt{2}$

b

i $a=3$

ii $c=6$

c

i $a=\dfrac{4\sqrt{3}}{3}$

ii $c=\dfrac{8\sqrt{3}}{3}$

7

a
i $b = 19$ ii $c = 19\sqrt{2}$

b
i $b = \dfrac{3}{4}$ ii $c = \dfrac{3\sqrt{2}}{4}$

c
i $b = 5\sqrt{2}$ ii $c = 10$

d
i $b = 4\sqrt{3}$ ii $c = 8$

e
i $b = 3\sqrt{3}$ ii $c = 6$

f
i $b = 9\sqrt{6}$ ii $c = 18\sqrt{2}$

8

a
i $a = 3$ ii $b = 3$

b
i $a = 7\sqrt{2}$ ii $b = 7\sqrt{2}$

c
i $a = 2\sqrt{15}$ ii $b = \sqrt{30}$

d
i $a = \dfrac{9\sqrt{6}}{2}$ ii $b = \dfrac{9\sqrt{6}}{2}$

e
i $a = 5$ ii $b = 5\sqrt{3}$

f
i $a = \dfrac{7}{2}$ ii $b = \dfrac{7\sqrt{3}}{2}$

9 $\dfrac{13}{2}$

10

a $p = 2\sqrt{3}$ units

b $q = 4\sqrt{3}$ units

c $r = 3$ units

d $s = 3\sqrt{3}$ units

e $\left(9\sqrt{3} + 3\right)$ units

11

a $148\sqrt{2}$ pixels

b 148 pixels

12

a $\dfrac{15\sqrt{2}}{2}$ units

b $30\sqrt{2}$ units

13 $\dfrac{16\sqrt{3}}{9}\ \mathrm{m}^2$

14 $36\ \mathrm{cm}$

Additional questions

15 $\sqrt{2}$ cm

16

a 1

b 45°

17

a $\sqrt{3}$

b 60°

18

a 30°

b $(3+\sqrt{3})$ cm

19

a $x=\dfrac{\sqrt{30}}{2}$

b $h=1$

c $x=2$

d $x=4\sqrt{6},\ y=4\sqrt{3}$

e $u=8,\ v=4\sqrt{3}$

f $a=\dfrac{9\sqrt{3}}{2},\ b=\dfrac{9}{2}$

g $x=7\sqrt{2},\ y=14\sqrt{2}$

h $a=9\sqrt{2},\ b=6\sqrt{3},\ c=12\sqrt{3}$

20

a 45°

b 60°

21

a 60°

b 30°

c x units

d $x\sqrt{3}$ units

e i False. The hypotenuse of a $30^\circ-60^\circ-90^\circ$ triangle is twice the length of the shorter leg.

ii True

iii False. The longer leg of a $30^\circ-60^\circ-90^\circ$ triangle is $\sqrt{3}$ times the length of the shorter leg.

iv True

23

	Statements	Reasons
	To prove: $c = a\sqrt{2}$	
1.	$\triangle ABC$ is a $45^\circ - 45^\circ - 90^\circ$ triangle	Given
2.	$a^2 + a^2 = c^2$	Pythagorean theorem
3.	$2a^2 = c^2$	Simplify
4.	$a\sqrt{2} = c$	Square root of both sides
5.	$c = a\sqrt{2}$	Symmetric property of equality

24 Yes, Zeki is correct. All isosceles right triangles are similar. An isosceles triangle has two equal interior angle measures, and so if it is also a right triangle the two congruent angles must have a measure of 45°. So any isosceles right triangle has angles $90^\circ, 45^\circ, 45^\circ$. This means that, by the Angle-Angle theorem, all isosceles right triangles are similar.

25 A $30^\circ - 60^\circ - 90^\circ$ triangle can be used to solve the problem. The hypotenuse of a $30^\circ - 60^\circ - 90^\circ$ triangle is twice the length of the shorter leg. The hypotenuse of the bookcase is twice the length of the base so we know the bookcase is a $30^\circ - 60^\circ - 90^\circ$ triangle. The angle formed between the base and the hypotenuse is 60°.